Scheduling

Backbone of multiprogramming. By managing multiple processes concurrently, the OS can interleave execution, wait time ↓ system throughput ↑

Process: A job that is being executed.
(The entity currently assigned to processor)
Prog code + all info for execution.

States of a Process

New: Just created + undergoing initialization by OS.

Ready: Initialized, waiting in ready queue to be assigned to proc.

Running: Actively being executed by CPU.

Waiting: Temporarily suspended as waiting for system resource

Halted: Completed + marked for removal from Mem.

Process Control Block (PCB)

↗ Maintained by OS for every process

↓

a data structure ⇒ holds critical info for process managmnt & Scheduling.

Identifier: Unique ID to distinguish the process.

State: Current state (new . _ . halted).

priority: relative importance of the process
↳ influencing scheduling.

Program Counter: Address of next instr to execute.

Memory Pointers: process location in memory.
(start, end addrs)

Context Data: Current contents of CPU registers. → reg val

(saved & stored during context switches)

I/O status info: Track → i/o req
- assigned dev
- file handles

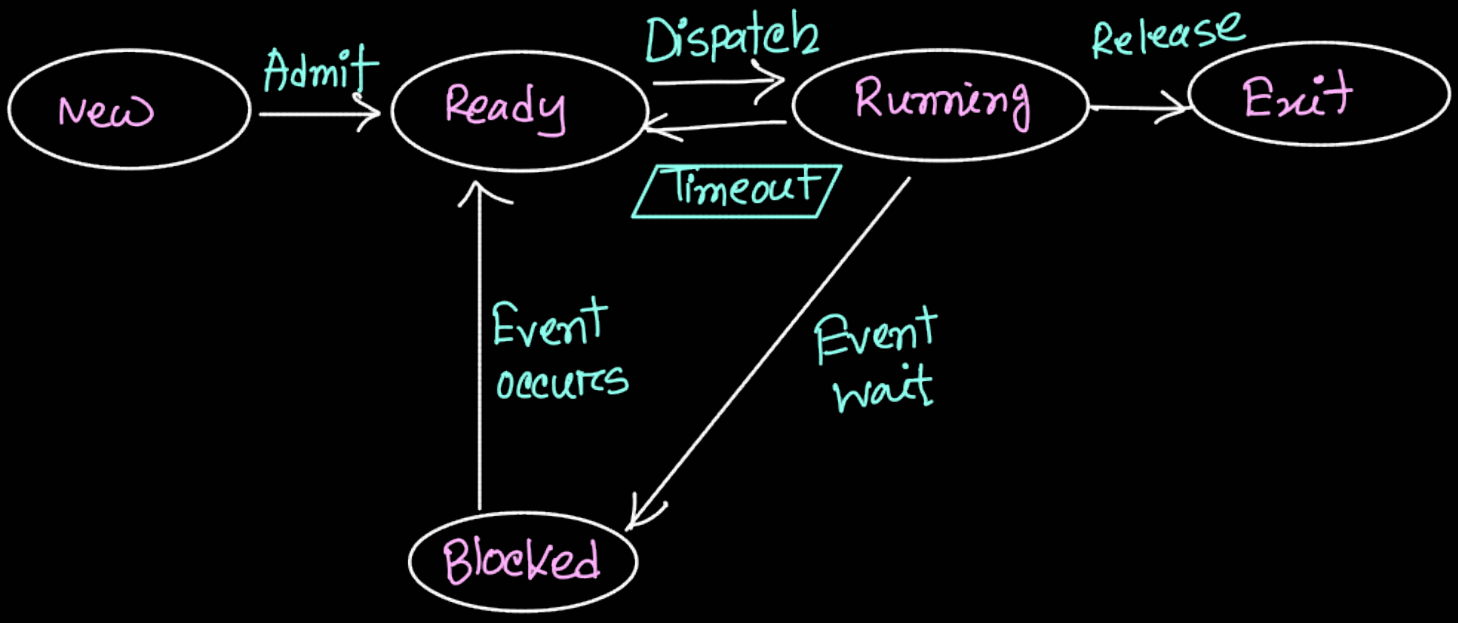
Accounting Info: → biling purpose
- CPU usage time
- elapsed clock time

+

other resource utilization Mat.

When a process is created, a blank PCB is set up.

initialization → move to ready state (wait for CPU Allocation)



Types of Scheduling

Long Term : Decides which progs or jobs are admitted into sys.

Function: Control Degree of Multiprog by regulating # of processes in mem

Operation in Batch sys:

→ Jobs queued on Disk until resources available

→ long term scheduler converts → Processes

places into → ready queue.

Interactive sys: For time sharing,

User logs in → process reqs immediately generated

OS accepts user connection till sys capacity.

Medium term scheduling: Manages swapping of processes in & out of M.M.

Function: helps maintain an optimal degree of Mult prog.

by removing processes (swap out) → when mem load.

reintroduce (swap in) → as resources become available.

N.B.: For sys without virtual mem, decisions influenced by mem requirement of each proces.

Short term: Select which of the ready process will be next executed by CPU.

Function: Acts as dispatcher, making rapid decision to maintain CPU efficiency.

Context switching:

⇒ A process in running state

interrupted (i/o req / service call / timeout)

OS saves context (reg val & PC) in PCB

→ Short term scheduler picks another process from ready queue & instructs CPU to load saved context resuming execution.

Scheduling Algo:

Round-Robin: processes given fixed time slice in cycling order.

priority Based: process with higher priority chosen over lower ones.

I/O Scheduling: (→ handles device operation) Manages pending I/O req across various devices.

(Func) Determines the order in which I/O operations are performed.

Queues:

- Each i/o device maintains own queue

- I/O requesting process → placed in → corresponding i/o q
- I/O operation complete, process is moved back to ready q for further execution.

Process Transition & OS Control:

1. Process Req & Creation:

→ new job enters sys & placed in long term q.

→ OS creates a PCB & transitions the process to ready state.

2. Dispatching:

→ Short term scheduler selects a process from ready q.

→ CPU loads the process' context from its PCB & begins execution.

3. Interrupts & System calls:

(i) A process may → req I/O
or
→ encounters error

(ii) An interrupt generated → CPU to switch to OS's interrupt or service call handlers.

(iii) OS saves the current process context & initiates I/O operation.

4. I/O completion & scheduling:

→ I/O op complete → associated process removed from I/O q to ready q.

→ ST scheduler selects it for CPU execution

5. Termination:

process complete/ aborted → marked as halted & removed from sys.

